

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

LABORATORY REPORT SHEET

Name : HIMANSHU SHARMA Subject : PHY 461
Name of Lab Partner : GAURAV KUMAR Roll No. : 230474
Section : Instructor : PROF. ZAKIR
Date of Experiment : 11/08/26
Date of Submission : 18/08/26
Experiment : PHOTOCONDUCTIVITY
Remarks by the Instructor :

OBJECTIVES:

- (i) Measure the photocurrent I_{ph} as a function of voltage V at a constant irradiance Φ .
- (ii) Measure the photocurrent I_{ph} as a function of irradiance Φ at a constant voltage V .

INTRODUCTION & THEORY

Photoconductivity is the effect of increasing electrical conductivity in a solid due to light absorption. When the so called internal photo-effect takes place, the energy absorbed enables the transition of activator electrons into the conduction bands and the charge exchange of traps with holes being created in the valence band. Therefore, the number of charge carriers in the crystal lattice and, as a result, the conductivity is enhanced. The change in conductivity is expressed as:

$$\Delta \sigma = \Delta p \cdot e \cdot \mu_p + \Delta n \cdot e \cdot \mu_n \quad \text{--- (1)}$$

where:

- e : elementary charge
- Δp : change of hole conc.
- Δn : change of e^- conc.
- μ_p : mobility of holes
- μ_n : mobility of e^-

When voltage V is applied, the photocurrent is:

$$I_{ph} = (A/d) \Delta \sigma \cdot V \quad \text{--- (2)}$$

where A is the cross section of the current path and d is the distance b/w the electrodes. (Fig 1).

Semiconductor resistors whose resistance depends on the irradiance (photoconductive cell) are based on this principle. They have The semiconductor material most commonly used are cadmium compounds, particularly CdS. In this experiment, a CdS photoresistor is exposed to light from a lamp. The irradiance and at the position of the photoresistor being varied by means of two polarization filters which are placed one behind the other. If the two polarization ~~plate~~ planes of the filter are rotated against each other by an angle α , the irradiance is

$$\Phi = \Phi_0 \cdot D \cdot \cos^2 \alpha$$

Φ_0 = irradiance without polarization

D = transparency when polarization planes are //

The photo current is studied as a function of the voltage applied to the photoresistor at a constant irradiance (current-voltage characteristics) and as a function of the irradiance at a constant voltage (current-irradiance characteristics)

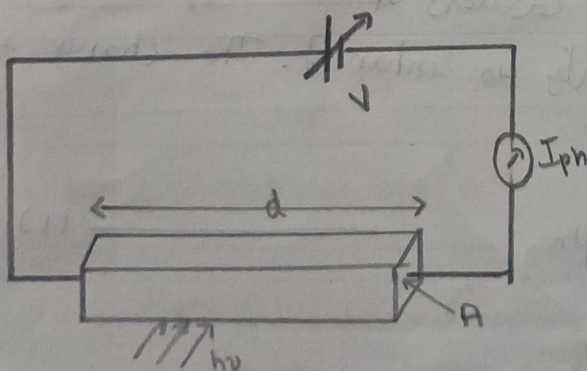
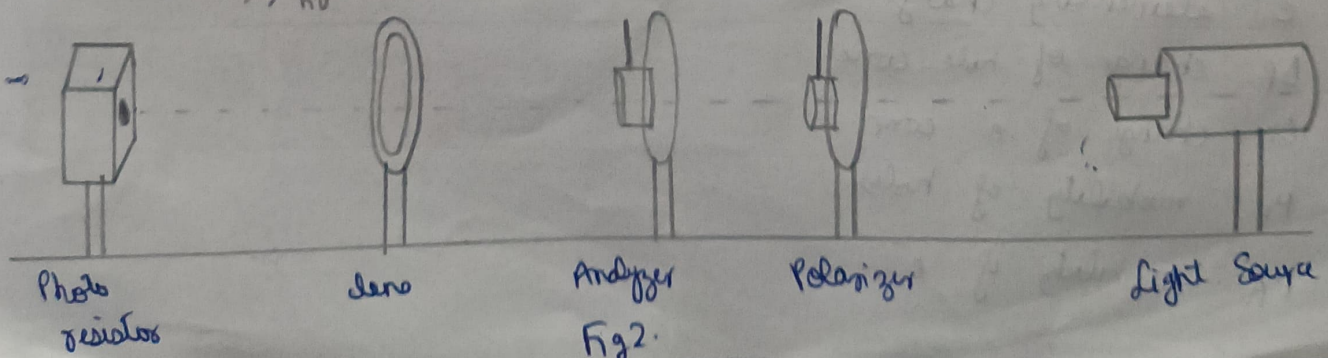


Fig1

Fig1 - Circuit Diagram

Fig2 - Schematic Diagram



OBSERVATIONS

(a) Measurement of the photocurrent I_{ph} as a function of voltage V at a constant irradiance Φ

Voltage (V)	I_{ph} (0°) (mA)	I_{ph} (30°) (mA)	I_{ph} (60°) (mA)	I_{ph} (90°) (mA)
16	0.591	0.528	0.660	0.992
14	0.524	0.472	0.613	0.855
12	0.468	0.402	0.535	0.738
10	0.401	0.331	0.467	0.665
8	0.327	0.262	0.381	0.551
6	0.261	0.232	0.308	0.433
4	0.188	0.176	0.220	0.300
2	0.128	0.124	0.143	0.178

(b) Measuring the photocurrent I_{ph} as a function of the irradiance Φ at a constant voltage V :

α (degree)	I_{ph} (40V) (mA)	I_{ph} (8V) (mA)	I_{ph} (1V) (mA)	α (degree)	I_{ph} (40V) (mA)	I_{ph} (8V) (mA)	I_{ph} (1V) (mA)
360	0.209	0.334	0.104	270	0.282	0.504	0.130
350	0.232	0.371	0.112	260	0.267	0.472	0.123
340	0.247	0.404	0.118	250	0.248	0.426	0.118
330	0.264	0.435	0.124	240	0.227	0.391	0.109
320	0.280	0.485	0.128	230	0.208	0.350	0.104
310	0.288	0.501	0.131	220	0.188	0.308	0.099
300	0.296	0.524	0.134	210	0.180	0.304	0.096
290	0.299	0.526	0.133	200	0.182	0.301	0.096
280	0.292	0.522	0.133	190	0.191	0.325	0.099

α (degree)	$I_{ph}(40V)$ (mA)	$I_{ph}(8V)$ (mA)	$I_{ph}(1V)$ (mA)		α (degree)	$I_{ph}(40V)$ (mA)	$I_{ph}(8V)$ (mA)	$I_{ph}(1V)$ (mA)
180	0.208	0.358	0.103		180 90	0.306	0.551	0.128
170	0.232	0.392	0.110		170 80	0.299	0.520	0.123
160	0.252	0.445	0.114		160 70	0.274	0.477	0.120
150	0.280	0.487	0.120		150 60	0.247	0.435	0.112
140	0.302	0.519	0.123		140 50	0.222	0.385	0.105
130 130	0.308	0.545	0.128		40	0.205	0.346	0.099
120 120	0.315	0.570	0.132		30	0.196	0.305	0.096
110 110	0.316	0.583	0.132		20	0.196	0.298	0.096
100	0.313	0.575	0.129		10	0.203	0.312	0.098
					0	0.216	0.333	0.102

(initially we had to take only three observations, but we were instructed to take the 4th observation {const volt, vary angle} later on so the 4th voltage observation data is present separately)

α (degree)	360	350	340	330	320	310	300	290	280	270
$I_{ph}(16V)$ (mA)	0.594	0.645	0.715	0.814	0.826	0.870	0.887	0.836	0.814	0.767
α (degree)	260	250	240	230	220	210	200	190	180	170
$I_{ph}(16V)$ (mA)	0.696	0.638	0.598	0.556	0.510	0.491	0.494	0.530	0.580	0.624
α (degree)	160	150	140	130	120	110	100	90	80	70
$I_{ph}(16V)$ (mA)	0.690	0.780	0.850	0.951	0.984	0.990	0.991	0.984	0.872	0.838
α (degree)	60	50	40	30	20	10	0			
$I_{ph}(16V)$ (mA)	0.658	0.584	0.552	0.536	0.542	0.573	0.598			

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DATA ANALYSIS & CALCULATIONS and ERROR ANALYSIS

least count of power supply = 0.1V

least count of multimeter = 0.001 mA

least count of polarizer angle measurements = 2.5 degrees

(a) I_{ph} as a function of V at constant Φ

$$I_{ph} = (A/d) \Delta\sigma \cdot V \quad \text{where } A = 2.5 \times 10^{-5} \text{ cm}^2$$

$$d = 2 \times 10^{-1} \text{ cm}$$

Due to fluctuations, we are getting intercept in the best fit line which we fit using least squares method. In order to get rid of the intercept, we can do one of these two things:

(i) simply ignore the intercept

(ii) ~~set (0,0) to be a~~ Fit $y = mx$ rather than $y = mx + c$.

we will demonstrate calculations via both these methods.

(i) Fit for $y = mx + c$

α (degrees)	slope (m) (mA/V)	$\sigma_m = \sqrt{\text{Cov}_{mm}}$ (mA/V)	$\Delta\sigma = m d/A$ (mA/V ² cm ²)	$\sigma_{\Delta\sigma} = \sigma_m d/A$ (mA/V ² cm ²)
0	0.0334	0.000395	0.2674	0.003988
30	0.0291	0.001081	0.2327	0.008647
60	0.0378	0.000919	0.3024	0.007348
90	0.0569	0.001288	0.4540	0.007348 0.010304

Note: Fitting for $y = mx + c$ ~~assumes that~~ and then ignoring the intercept means shifting all the points by the intercept. This means we are operating with the assumption that the instrument has a constant ~~noise~~ in photo current at 0 voltage due to reasons like background photons.

(ii) Fitting for $y = mx$

α (degrees)	slope (m) (mA/V)	$\sigma_m = \sqrt{\text{Cov}_{mm}}$ (mA/V)	$\Delta\sigma = m d/A$ (mA/V cm)	$\sigma_{\Delta\sigma} = \sigma_m d/A$ (mA/V cm)
0	0.0387	0.001035	0.3099	0.008277
30	0.0338	0.001024	0.2769	0.008194
60	0.0445	0.001341	0.3558	0.010729
90	0.0638	0.001439	0.5103	0.011513

Note: Fitting for $y = mx$ means assuming the deviation of the points to be the individual statistical error of the observation.

Note: Along with the statistical errors, there exists instrumental uncertainty due to the least count of measurements. It is safe to assume that the two RVs (stat errors and instrumental uncertainty) are independent. Hence, we can just use $\sigma_{\text{tot}} = \sqrt{\sigma_1^2 + \sigma_2^2}$

Systematic Errors

$$\frac{\delta(\Delta\sigma)}{(\Delta\sigma)} = \left| \frac{\delta m}{m} \right| + \left| \frac{\delta d}{d} \right| + \left| \frac{\delta A}{A} \right|$$

$$= \left| \frac{\delta V}{V} \right| + \left| \frac{\delta I}{I} \right|$$

{ we'll take δd and δA to be 0
as they are provided to us }

now $V_{\text{meas}} = (1 + \delta V/V) V_{\text{true}}$

$I_{\text{meas}} = (1 + \delta I/I) I_{\text{true}}$

$\Rightarrow m_{\text{meas}} = \left(\frac{1 + \delta I/I}{1 + \delta V/V} \right) m_{\text{true}}$

Note: For the sake of simplicity, we assume that the systematic errors due to least count of the instruments are accounted for in the statistical errors

(b) I_{ph} as a function of irradiance Φ at constant V

we plot the experimental observations of photocurrent (I_{ph}) (mA) v/s angle α (degrees). ~~The behaviour~~ The expected behaviour is that of a cosine curve, so we fit according to:

$$I_{ph} = A \cos^2(\alpha - \phi) + C \quad \text{where } A, \phi, C \text{ are fit params.}$$

Note: ϕ (phase difference) tells us about the angular misalignment in axis of polarizer and analyzer.

Note: Non-zero C tells us that even when analyzer and polarizer are orthogonal, there exists some current due to imperfections in instruments & background photons.

Voltage (V)	Offset (C) (mA)	$\sigma_c = \sqrt{Cov_{cc}}$ (mA)	Amplitude (A) (mA)	$\sigma_A = \sqrt{Cov_{AA}}$ (mA)	Phase (ϕ) (degree)	$\sigma_\phi = \sqrt{Cov_{\phi\phi}}$ (deg)
1V	0.0971	0.000514	0.0368	0.000849	-66.34	0.6500
4V	0.1914	0.002874	0.1206	0.004752	-66.79	1.1276
8V	0.3044	0.006781	0.2511	0.011214	-68.16	1.2810
16V	0.4975	0.017854	0.4352	0.029513	-65.29	1.9350

now, $\Phi = \Phi_0 \cdot D \cdot \cos^2(\alpha - \phi)$, $I_{ph} = m_2 \Phi_{norm} + C_2$

we will plot v/s normalized irradiance, which is $\Phi / \Phi_0 D = \Phi / \Phi_{max}$

Voltage (V)	m_2 (slope)	$\sigma_{m_2} = \sqrt{Cov_{m_2}}$	C_2 (intercept)	$\sigma_{C_2} = \sqrt{Cov_{C_2}}$
1V	0.0368	0.000837	0.0971	0.000506
4V	0.1206	0.004682	0.1914	0.002831
8V	0.2511	0.011049	0.3044	0.006679
16V	0.4352	0.029078	0.4975	0.017584

This is an almost linear fit, but variance increases as voltage increases, or as Φ_{norm} increases.

Results

① Change in conductivity (photoconductivity) of CdS photodiode is

- (a) For $\alpha = 0^\circ$: $(0.2674 \pm 0.0040) \text{ mA V}^{-1} \text{ cm}^{-1} = (2.674 \pm 0.040) \times 10^{-2} \text{ S m}^{-1}$
 (b) For $\alpha = 30^\circ$: $(2.327 \pm 0.087) \times 10^{-2} \text{ S m}^{-1}$ (3.739% error)
 (c) For $\alpha = 60^\circ$: $(3.024 \pm 0.074) \times 10^{-2} \text{ S m}^{-1}$ (2.447% error)
 (d) For $\alpha = 90^\circ$: $(4.549 \pm 0.103) \times 10^{-2} \text{ S m}^{-1}$ (2.264% error)

- ② Photocurrent varies sinusoidally with angle b/w polarizer and analyzer
- ③ phase difference (difference in the alignment of axis of polarizer & analyzer) is ~ 66.645 degrees.
- ④ Photocurrent increases linearly with irradiance in the range of observation

Answers to Questions

Q1) State the difference b/w donor or acceptor dominated photoconductivity, and photoconductivity of intrinsic semiconductor.

Ans - Intrinsic photoconductivity occurs when the incident photons have energy equal to, or greater than, the bandgap energy and the photons excite the electrons from the valence band to conduction band. This leads to formation of free e^- and free holes (e^- -hole pair) which contribute to conductivity.

On the other hand, in doped semiconductors, when the energy of photon is less than the bandgap energy, the photons excite from impurity levels to the nearest energy level when $h\nu > E_a, E_d$. This process generates only majority carriers (either e^- from donors to the conduction band, or holes from acceptors to the valence band) rather than e^- -hole pairs.

Q2) How will you modify this experiment so that it allows measurement of donor/acceptor energy of a doped semiconductor and the band gap of an intrinsic semiconductor?

Ans - We measure the photocurrent as a function of photon energy ($h\nu$) or wavelength using a variable light source. By sweeping the photon energy in the low energy region, onset of photocurrent will indicate ionization energy.

Q3) How does the lifetime of photo-excited carriers affect the response of a photo-resistor? What factors limit the lifetime? Can you suggest a ~~meas~~ method to measure carrier lifetime?

Ans - Longer lifetime the carrier take longer to recombine when light is turned off, decreasing the response time. It also leads to greater photocurrent due to higher steady state concentration of photo-generated carriers. Various forms of recombinations like radiative (band to band), Auger (energy transfer to 3rd carrier), SRH (mediated by defect or trap states), surface recombinations limit lifetime. To measure lifetime, we can excite the semiconductor by a short, intense pulse of light and then check the photocurrent response hence plot photoconductivity vs time ($e^{-t/\tau}$).

Q4) What are excitons? Do they play a role in photoconductivity?

Ans - Exciton is a neutral quasiparticle comprised of an e^- and a hole bound by Coulomb interaction (attractive). Excitons do not directly contribute to photoconductivity. However if an exciton gains enough thermal energy, interacts with a defect, or is subjected to a strong electric field, it can dissociate into free e^- and hole and these free carriers contribute to conductivity.